

Generation Cardinality as a Forced Role-Occupancy Endpoint

Rank-matching closure over the open-generation Standard Model carrier architecture

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Abstract

This manuscript proves an open-generation role-occupancy theorem for the Standard Model carrier architecture. The target is not a numerical prediction of masses, CKM entries, PMNS entries, mixing angles, CP phases, or oscillation probabilities. It is the internal generation-cardinality question: how many mutually compatible generation-role occupancies can close when representation response, charge-lattice compatibility, weak response, mass-response slots, flavor-transition standing, rephasing quotient, CP/holonomy residue, boundary reconstruction, no-selector discipline, and role-occupancy closure are imposed simultaneously.

The proof is a rank-matching argument. A closed candidate with generation cardinality n has a quotient-normalized transition graph Γ_n . Pairwise co-comparability forces $\Gamma_n \simeq K_n$. Rephasing quotienting identifies independent non-removable transition residues with the cycle-space quotient, so the raw circuit rank is

$$\text{rank } \Omega_{\text{raw}}(K_n) = \beta_1(K_n) = \frac{(n-1)(n-2)}{2}.$$

The decisive target theorem is that the same-scope Standard Model generation target requires and licenses exactly one primitive CP/holonomy orientation-residue register:

$$\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1.$$

This is not inferred from the singular name of a carrier type. It is proved by a local instantiation of fixed-domain exhaustion and same-scope operator closure in the generation-holonomy arena. A zero-rank reading would collapse the CP/holonomy-residue carrier into phase-coordinate skin or flavor-transition bookkeeping. A second independent primitive residue would induce a second same-scope standing-relevant orientation-residue operator; fixed-domain exhaustion and operator closure classify every such second operator as bookkeeping-equivalent, registered, inadmissible, extension, target update, burden, failure, or hostile branch, not a second closed primitive register.

Closed generation certification therefore implies

$$\text{rank } \Omega_{\text{live}}(K_n) = \text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1,$$

and hence $\beta_1(K_n) = 1$, forcing $n = 3$. The constructive direction is supplied by the minimal closed transition triangle, with the three role positions unique up to lawful role-incidence permutation. The proof uses no empirical family-count selector, no CKM/PMNS dimension selector, no charged-lepton-name selector, no fitted texture, and no fourth-generation non-observation argument. It does not show that no target-updating or extension theory can contain additional fermion families; it shows that such additions do not close the same-scope Standard Model generation-role certificate. It exposes a hostile branch: a valid same-scope certificate with a different primitive residue rank would falsify the theorem rather than being reclassified away.

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1 Reader orientation

1.1 Problem statement

The Standard Model carrier arc separates primitive carrier standing from derivative skins, measurements, numerical values, routes, projections, matrix coordinates, and model surfaces. In the flavor sector this discipline already blocks generation labels, field duplication, matrix enlargement, mass-value addition, and anomaly repetition from becoming primitive generation standing. The remaining question is stronger: whether the carrier architecture itself forces the number of generation-role occupancies.

The theorem proved here is

$$\text{ClosedGenCert}(n) \iff n = 3. \quad (1)$$

The certificate is evaluated in an open-generation arena D_{gen} . It is not evaluated in the fixed-envelope arenas $D_{\text{flavor}}^{N_g}$ or $D_{\text{SM}}^{N_g}$, where the family-cardinality load is already governed.

1.2 Proof overview

The proof has four steps.

1. A closed generation-role certificate forces a complete co-comparability graph $\Gamma_n \simeq K_n$.
2. Rephasing quotienting identifies independent transition-orientation residues with the cycle-space quotient, whose rank for K_n is $(n-1)(n-2)/2$.
3. Fixed-domain exhaustion and same-scope operator closure prove that the Standard Model generation target has primitive holonomy-residue rank one.
4. Rank matching gives $(n-1)(n-2)/2 = 1$, so $n = 3$; the triangular assignment supplies the positive certificate.

1.3 Nonclaims

The theorem does not derive numerical masses, Yukawa eigenvalues, CKM entries, PMNS entries, mixing angles, CP phase values, neutrino ordering, or oscillation probabilities. It does not use observed charged-lepton labels, observed neutrino flavor count, LEP data, or experimental non-observation of fourth generations. It also does not deny BSM physics. A target-changing or extension claim remains scientifically admissible as an extension, burden, or target update; it simply does not close as a same-scope Standard Model generation-role certificate.

1.4 Falsifiability

A valid hostile certificate must preserve the same carrier target, defeat registered status and quotient collapse, avoid hidden selectors and hidden load, and close compatible role occupancy with primitive residue rank different from one. Such a certificate would falsify the theorem. The classifier is not permitted to absorb it by vocabulary.

2 Theorem-level source reception

2.1 Critical source theorem ledger

The following table records the received theorem payloads used in the proof. Each source is used only at the stated output.

Table 1: Theorem-level dependency ledger.

Source	Received theorem / result	Payload used here
Role-Occupancy Closure	Definitions 3.5–3.18; Theorem 7.4; Corollary 7.5	Forced target role set, eligibility EC, joint compatibility JC, compatible role-covering assignment, and cardinality equality as consequence rather than proof ground.
Role-Cardinality Closure	Theorems 6.1, 7.2, 7.4; Corollaries 7.5, 7.7; Theorem 8.3	Forced role-set discipline, no cardinality backfitting, mismatch disposition, closure routes for over-/under-residue.
Generation Count and Residual Parameters	Propositions 9.2, 9.3, Lemma 9.4, Theorem 9.8, Theorem 12.2	Generation count is not a presentation coordinate; no same-scope generation selector; fixed-boundary reading in older SM scope; no hidden generation binder.
Generation-Count Bounds	Lemma 1, Lemma 2, Proposition 1	Unbounded multiplicity is illicit; bounded unforced multiplicity collapses; generation count is not mechanically derived by residual display alone.
Mixing-Angle and Phase Standing	Theorems 5.2, 5.3, 5.5, 6.3, 7.2, 9.4, 11.6, 17.3, 18.2	No mixing from one carrier; independent-carrier precondition; common interface; canonical mixing carrier; rephasing quotient; CP residue criterion; raw parameter count is not role-rank; conditional quotient count.
CNF7	Theorems 6.2, 7.2, 8.2, 11.1, 16.2, 18.2, 21.2	Independent-carrier and interface preconditions; mixing standing attaches to canonical carrier; rephasing quotient; CP residue standing; phase-holonomy forcing; PMNS boundary reconstruction.

Source	Received theorem / result	Payload used here
CP-like Phases as Holonomy Classes	Lemma 1, Theorem 1, Lemma 2, Lemma 3, Theorem 2	Removable phase deletion; non-removable phases are holonomy classes; holonomy-class status is distinguished from raw phase-coordinate status; additional independent holonomy classes require target-load increase if asserted as primitive. The at-most-one primitive register theorem is proved in this manuscript, not imported as a cardinality premise.
L1 Paper 8	Theorem 1; Corollaries 1–2	Unitary mixing saturation; matrix enlargement does not generate primitive mixing freedom under fixed load.
L1 Paper 9	Theorem 1; Corollaries 1–2	Rephasing redundancy exhaustion; surplus phase freedom is removable or load-changing.
Neutrino Oscillation Reconstruction	Theorems 4.14, 4.18, 5.5, 5.6, 6.3, 6.6, 8.4, 9.1, 10.3, 13.1	PMNS reconstruction quotient; hidden witness collapse; no global flavor transport; boundary trace not interior identity; flavor invariance iff commutation; rival exhaustion.
NNR6	Theorems 4.1, 6.1, 7.1, 8.1, 10.1	PMNS-like boundary carrier only; no hidden flavor transport from ratio interval; no mixing-angle value; CP phase remains deferred; shape restriction is not mixing extraction.
CLD	Theorems 4.1, 4.2, 5.1, 6.1, 9.1, 14.1	Cardinality readiness requirement; current-family residues are non-promotable; role/scalar cardinality separation; quotient-normalized cardinality; reconstruction permission audit.
SM-5 Flavor/Mixing/Generation	Theorems 5.1–5.5, 7.2, 8.2, 10.2, 12.1–12.3, 13.5, 17.1–17.6	Flavor-transition and CP/holonomy carriers; fixed generation governance; rephasing and CP classification; PMNS boundary reconstruction; generation label/field duplication nonpromotion; fourth-generation branch; fixed-envelope nonclaim.

Source	Received theorem / result	Payload used here
SM-6 Closure	Target-update, registered-status, hostile-branch, and no-leakage discipline	A valid hostile certificate falsifies rather than being absorbed; registered statuses do not become primitive by utility or empirical salience.
Fixed-Domain Exhaustion	Definition 0.1; Theorems 7.1, 7.2, 7.3, 7.4, 11.1, 12.1; Corollary 7.1	Same-scope variation has only boundary, standing-partition, quotient, and continuation loci; no second classifier; no fourth role; operator identity is invariant under lawful redescription.
Same-Scope Operator Closure	Lemma 2.9; Theorems 3.10, 4.5, 5.13, 6.14, 7.1; Corollaries 7.2–7.4	Auxiliary-datum dichotomy; no same-scope totalization; every same-scope operator is bookkeeping-equivalent or inadmissible; no emergent operator by composition, iteration, diagonalization, or foundation swap.
Impossibility Suite	Kernel compression: no generators, no carriers, no repairs	Compressed impossibility spine for no generation of legitimacy, no cross-domain transfer, and no post-hoc repair of failed same-scope residue certification.

2.2 Circularity audit for the CP-holonomy source

The CP-like holonomy source is used for three non-cardinality payloads: removable-phase deletion, holonomy-class status for non-removable phase-like residue, and the rule that an additional independent holonomy class is a target-load increase if asserted as primitive. The manuscript does not import the statement that the Standard Model has three generations, does not import a 3×3 CKM or PMNS dimension, and does not use empirical CP data as a selector. The at-most-one primitive register result is proved below from fixed-domain exhaustion, residue-to-operator conversion, and same-scope operator closure. Thus the CP-holonomy source supplies residue typing, not the generation cardinality theorem.

3 Open-generation arena and role certificate

Definition 3.1 (Open-generation arena). *The open-generation arena is*

$$D_{\text{gen}} = (D_{\text{SM}}^{\text{open}}, C_{\text{gauge}}, C_{\text{matter}}, C_{\text{CCR}}^{\text{prim}}, H_{\text{br/fix}}, C_{\text{flavor-transition}}, C_{\text{CP/hol}}, \text{GenAdm}, \text{NoSelector}, \text{NoHiddenLoad}). \quad (2)$$

It preserves the Standard Model carrier architecture but contains no fixed family-cardinality envelope.

Definition 3.2 (Candidate cardinality arena). *For $n \in \mathbb{N}$, $D_{\text{gen}}(n)$ is D_{gen} plus a candidate presentation of n role-distinct generation residues before final role-occupancy closure.*

Theorem 3.3 (Open-arena lifting of the CP/holonomy carrier). *The carrier role $C_{\text{CP/hol}}$ lifts from the fixed-envelope flavor arena into D_{gen} as a carrier-role requirement without importing fixed generation cardinality. The lifted payload is the distinction between primitive holonomy-residue standing and registered phase-coordinate status; the number of live primitive residue registers is determined internally by D_{gen} .*

Proof. SM-5 receives the CP/holonomy-residue carrier as part of the primitive flavor-side package while explicitly treating CKM entries, PMNS entries, raw phase coordinates, basis choices, rephasings, and generation labels as nonprimitive statuses. That payload does not require a fixed value of N_g ; it requires only that the open-generation target retain the same flavor-transition carrier role and the same no-promotion discipline for raw coordinate phases. In D_{gen} , those carrier-role and status-typing conditions are held fixed while the candidate cardinality is left open. Therefore the CP/holonomy carrier lifts as a role requirement, whereas its live primitive rank is not inherited from the fixed-envelope arena. The rank is proved in Section 10. $\square$

Definition 3.4 (Generation role). *A candidate r satisfies $\text{GenRoleCert}(r)$ when*

$$\begin{aligned} \text{GenRoleCert}(r) := & R(r) \wedge Q(r) \wedge W(r) \wedge M(r) \wedge F(r) \\ & \wedge B(r) \wedge \Omega(r) \wedge S(r) \wedge J(r), \end{aligned} \quad (3)$$

where R is representation-response eligibility, Q charge-lattice compatibility, W weak-chiral compatibility, M mass-response-slot compatibility, F flavor-transition participation, B boundary-reconstruction compatibility, Ω CP/holonomy-residue compatibility, S no-selector compliance, and J joint compatibility with the other occupied roles.

Definition 3.5 (Presentation quotient). *Candidates are equivalent, $x \sim_{\text{role}} y$, if their difference is only a field-copy label, family-label permutation, basis choice, rephasing convention, matrix-coordinate presentation, mass-value display, numerical fit, detector route, or observed-family name. Define*

$$\overline{\text{Residue}}_n := \text{Residue}(D_{\text{gen}}(n)) / \sim_{\text{role}}. \quad (4)$$

Lemma 3.6 (No generation by presentation). *Field-copy multiplicity, matrix-size enlargement, mass-value addition, anomaly repetition, and family-label syntax do not create role-distinct generation residues.*

Proof. This is the direct open-generation analogue of the SM-5 generation-label and field-duplication nonpromotion theorems, combined with CLD quotient-normalization and role/scalar cardinality separation. A presentation change can be licensed as a representative, registered coordinate, burden, or target-update pressure. It becomes a role-distinct residue only by satisfying GenRoleCert after quotienting. Therefore presentation multiplicity alone is deleted before cardinality is evaluated. $\square$

4 Closed generation certificate

Definition 4.1 (Closed generation certificate). *ClosedGenCert(n) holds iff there exists a forced target role set $\text{RT}(D_{\text{gen}}, \text{GenerationRole})$, a live quotient-normalized residue $\overline{\text{Residue}}_n^{\text{live}}$, and a unique compatible role-covering assignment*

$$\sigma : \text{RT}(D_{\text{gen}}, \text{GenerationRole}) \rightarrow \overline{\text{Residue}}_n^{\text{live}} \quad (5)$$

modulo lawful role-incidence automorphism such that

$$\forall r \in \text{RT}, \quad \text{EC}(\sigma(r), r), \quad \text{JC}(\sigma) = 1, \quad (6)$$

with all certificate fields in `GenRoleCert` satisfied and with no hidden selector, hidden load, target update, or surplus primitive residue.

Remark 4.2. Cardinality equality is a consequence of this certificate, not a proof ground. The role-occupancy source explicitly blocks bare residue count, singleton residue, empirical fit, or backfilled target roles as realization certificates.

Lemma 4.3 (Certificate implies live transition rank match). *If $\text{ClosedGenCert}(n)$ holds, then the live quotient-normalized generation-transition residue rank must match the primitive target holonomy rank:*

$$\text{rank } \Omega_{\text{live}}(\Gamma_n) = \text{rank } \Omega_{\text{SM-gen}}^{\text{prim}}. \quad (7)$$

Proof. A closed certificate requires compatible occupation of every forced target role and no surplus primitive residue. If the live transition residue rank is smaller than the target rank, the assignment under-occupies the CP/holonomy-residue role. If it is larger, the assignment contains surplus independent primitive transition-circuit load. Role-occupancy closure permits neither under-residue nor over-residue inside a fixed target. Hence the live ranks must match. Raw phase-coordinate rank is not used in this lemma; raw rank enters only after live-circuit necessity is proved in Section 9. $\square$

5 No-selector theorem

Theorem 5.1 (No-selector theorem). *Let x be an empirical family surface, matrix dimension, field-copy display, mass table, fitted texture, or observed flavor label. If x lacks a role-set forcing certificate in D_{gen} , then x cannot function as a proof of generation cardinality:*

$$\text{EmpiricalFamilySurface}(x) \wedge \neg \text{RoleSetForcingCert}(x) \Rightarrow \neg \text{GenerationCardinalityProof}(x). \quad (8)$$

Proof. SM-1 blocks primitive standing from field symbols, matrix entries, numerical values, measurements, routes, projections, and model names without carrier certificates. SM-5 blocks generation-label, field-duplication, matrix-size, and mass-value promotion. CLD blocks current-family residue and scalar-output promotion without cardinality-readiness and quotient normalization. Therefore no empirical or presentation surface can select n before role-set forcing closes. $\square$

6 Transition graph normal form

6.1 Co-comparability graph

Definition 6.1 (Generation-transition graph). *For a candidate n -presentation, the generation-transition graph Γ_n has vertices given by role-distinct candidate generation residues after presentation quotient. An edge e_{ij} is present when r_i and r_j are directly co-comparable under the same matter/mass/weak/flavor-transition certificate.*

Definition 6.2 (Direct co-comparability). *Two role-distinct occupants r_i, r_j are directly co-comparable when the same certified architecture supplies all of the following without additional selector or target change:*

1. common representation-response comparison type;
2. common charge-lattice and weak-response comparison type;
3. common mass-response-slot comparison type;

4. *common flavor-transition target;*
5. *shared rephasing/basis quotient discipline;*
6. *shared boundary reconstruction discipline where neutrino-like routes are involved.*

Lemma 6.3 (Pairwise completeness is forced, not assumed). *If ClosedGenCert(n) holds, then every two role-distinct generation occupants are directly co-comparable.*

Proof. A closed generation certificate assigns all live residues to one target role architecture. Each role is not merely a matter field; it is a co-occupant of a common matter/mass/weak/flavor-transition structure. Suppose two live occupants r_i, r_j are not directly co-comparable. Then at least one certificate field has failed. If they do not share the representation, charge, weak, or mass-response comparison type, the pair cannot belong to the same generation-role target. If they share those fields but no flavor-transition comparison exists, $C_{\text{flavor}}(n)$ fails. If a hidden rule declares the pair incomparable while other pairs remain comparable, $C_{\text{nosel}}(n)$ fails. If the pair is connected only through a new mediator not already licensed by the target, hidden load has entered. If the system decomposes into disconnected components, the target is no longer the same-scope Standard Model flavor target. Each case contradicts ClosedGenCert(n). Hence pairwise co-comparability is forced. $\square$

Lemma 6.4 (Complete graph normal form). *If ClosedGenCert(n) holds, then*

$$\Gamma_n \simeq K_n. \quad (9)$$

Proof. By Lemma 6.3, every pair of distinct live residues is joined by a direct co-comparability edge. Thus the transition graph has all $\binom{n}{2}$ edges among its n vertices. This is precisely the complete graph K_n . The statement is a normal-form theorem: it follows from closure and no-selector discipline, not from a preference for complete graphs. $\square$

Definition 6.5 (Circuit rank). *For a connected finite graph G , the circuit rank is*

$$\beta_1(G) = |E(G)| - |V(G)| + 1. \quad (10)$$

For K_n , $n \geq 1$,

$$\beta_1(K_n) = \binom{n}{2} - n + 1 = \frac{(n-1)(n-2)}{2}. \quad (11)$$

Lemma 6.6 (First exact circuit rank). *For $n \geq 1$,*

$$\beta_1(K_n) = 1 \iff n = 3. \quad (12)$$

Proof. Using (11), $\beta_1(K_n) = 1$ gives

$$(n-1)(n-2) = 2,$$

so $n^2 - 3n = 0$, hence $n = 0$ or $n = 3$. In the nonempty connected case $n \geq 1$, the only solution is $n = 3$. $\square$

Table 2: Circuit rank of the closed co-comparability normal form.

n	Γ_n	$\beta_1(K_n)$	Cardinality role
0	none	none	no occupancy
1	K_1	0	standing without transition
2	K_2	0	transition edge without circuit
3	K_3	1	first exact holonomy circuit
4	K_4	3	surplus independent circuits
5	K_5	6	surplus independent circuits

7 Rephasing quotient and cycle-space residue

7.1 Edge phases are not primitive

The proof does not promote edge phases, matrix entries, or fitted mixing angles. Edge labels are transition coordinates. They become primitive-relevant only after all vertex-local rephasings, basis choices, and coordinate conventions are removed and a closed circuit residue remains.

Definition 7.1 (Edge-coordinate group and vertex rephasing). *For a transition graph Γ , let $C^1(\Gamma)$ denote the abelian group of edge phase-coordinate assignments and $C^0(\Gamma)$ the group of vertex-local phase choices. The rephasing boundary map*

$$d : C^0(\Gamma) \rightarrow C^1(\Gamma)$$

records the edge-coordinate change induced by rephasing vertex representatives.

Definition 7.2 (Cycle-space residue). *The rephasing-quotient residue is*

$$\Omega_{\text{res}}(\Gamma) = C^1(\Gamma)/dC^0(\Gamma), \quad (13)$$

restricted to closed transition-circuit classes. Equivalently, it is the first cohomological residue

$$\Omega_{\text{res}}(\Gamma) \cong H^1(\Gamma)$$

when only graph-level transition phases are being counted.

Lemma 7.3 (Rephasing deletion). *A phase assignment in $dC^0(\Gamma)$ is a rephasing skin and does not contribute to primitive CP/holonomy residue.*

Proof. A vertex-local rephasing changes representative coordinates while preserving the underlying role occupants and their transition standing. By rephasing redundancy exhaustion and the no-promotion discipline for coordinates, such a phase has no quotient-stable role. It is a registered phase/rephasing status, not primitive holonomy residue. $\square$

Lemma 7.4 (Cycle quotient theorem). *For a closed generation-transition graph Γ_n , the independent non-removable transition phase residues are exactly the cycle-space residues:*

$$\text{rank } \Omega_{\text{res}}(\Gamma_n) = \beta_1(\Gamma_n). \quad (14)$$

Proof. Edge coordinates alone are not invariant because endpoint representatives can be rephased. Quotienting by vertex-local rephasings removes all exact edge-coordinate changes. What remains are closed orientation sums around independent circuits. A spanning tree has no such residue after vertex rephasing; adding one independent edge beyond a tree creates exactly one independent closed circuit residue. Thus the number of independent quotient-stable residues is $|E| - |V| + 1 = \beta_1(\Gamma_n)$. $\square$

Corollary 7.5 (Complete-graph residue rank). *If $\text{ClosedGenCert}(n)$ holds, then*

$$\text{rank } \Omega_{\text{res}}(\Gamma_n) = \beta_1(K_n) = \frac{(n-1)(n-2)}{2}. \quad (15)$$

Proof. By Lemma 6.4, $\Gamma_n \simeq K_n$. Apply Lemma 7.4 and (11). $\square$

8 Generation-holonomy fixed-domain instantiation

8.1 Local standing-bearing carrier

The fixed-domain and same-scope operator theorems are used only after local instantiation. The local arena is

$$\mathcal{G}_{\text{hol}} = (X_{\text{gen}}, \Omega_{\text{op}}, \simeq_{\text{Adm}}, G_{\text{prim}}, T_{\text{reg}}). \quad (16)$$

Here X_{gen} is the standing-bearing generation-transition carrier: role-distinct generation occupants, certified pairwise co-comparability channels, and closed transition circuits after basis, rephasing, field-label, family-label, and matrix-coordinate quotienting. The class Ω_{op} contains candidate orientation-residue operators. The relation $\simeq_{\text{Adm}}$ is lawful redescription: basis change, rephasing, field relabeling, family relabeling, coordinate reparameterization, and matrix-skin change. The gate G_{prim} is the primitive-residue standing partition. The atlas T_{reg} contains raw phase coordinates, CKM/PMNS coordinates, boundary reconstruction, calibration, route, projection, extension, target update, burden, and failed-primitive statuses.

Table 3: Local instantiation of fixed-domain and same-scope operator hypotheses.

Abstract hypothesis	Generation-holonomy realization
H_1 : Fixed boundary signature	The Standard Model carrier target, flavor-transition role, CP/holonomy role, no-selector discipline, and role-occupancy grammar are held fixed.
H_2 : Evaluated fragment	Quotient-normalized closed transition-circuit classes in X_{gen} .
H_3 : Standing partition	Primitive residue, registered coordinate/boundary status, extension, target update, burden, or failed primitive.
H_4 : Lawful quotient	Basis change, rephasing, field-label, family-label, matrix-coordinate, and presentation redescription collected in $\simeq_{\text{Adm}}$.
H_5 : Admissible continuation	Transition-comparison continuation and role-occupancy continuation; no post-hoc repair of failed residue certification.

Abstract hypothesis	Generation-holonomy realization
H_6 : Role-typing carrier	The flavor-transition/CP-holonomy sector; generation governance is not a primitive carrier.
H_7 : No hidden selector	Empirical labels, matrix dimensions, textures, zero settings, fitted angles, and family names carry no primitive authority.
H_8 : No imported evaluator	An additional residue gate must be standing-fixed or it is imported data and fails same-scope admissibility.

Theorem 8.1 (Local fixed-domain instantiation). *The arena $\mathcal{G}_{\text{hol}}$ satisfies the hypotheses needed for fixed-domain exhaustion and same-scope operator closure.*

Proof. The verification is componentwise. The hypotheses listed in Table 3 are denoted H_1 through H_8 and are discharged as follows.

1. H_1 : *fixed boundary signature*. The admissible vocabulary is the SM carrier target together with the flavor-transition role, the CP/holonomy role, no-selector discipline, and the role-occupancy grammar. Introducing a new primitive phase gate, family selector, texture rule, or evaluator changes this boundary signature rather than refining the same target.
2. H_2 : *evaluated fragment*. The evaluated objects are quotient-normalized closed transition-circuit classes in X_{gen} . Field labels, family labels, matrix coordinates, basis choices, and rephasing representatives are removed before primitive rank is evaluated.
3. H_3 : *standing partition*. Every evaluated residue is assigned exactly one of the declared statuses: primitive residue, registered coordinate or boundary status, extension, target update, burden, or failed primitive. No intermediate status is used as hidden primitive authority.
4. H_4 : *lawful quotient*. The lawful quotient is $\simeq_{\text{Adm}}$, generated by basis change, rephasing, field-label redescription, family-label redescription, matrix-coordinate change, and presentation-equivalent rewriting. A proposed distinction inside an $\simeq_{\text{Adm}}$ -class is skin or selector import, not primitive residue.
5. H_5 : *admissible continuation*. The continuation structure is fixed by transition-comparison continuation and role-occupancy continuation. Failed residue certification is not repaired by later choosing a texture, gauge, convention, or post-hoc representative.
6. H_6 : *role-typing carrier*. The only role-typing carrier for the present question is the flavor-transition/CP-holonomy sector. Generation governance remains nonprimitive; it can govern labels and branch status but cannot supply an extra primitive residue register.
7. H_7 : *no hidden selector*. Empirical labels, matrix dimensions, zero entries, fitted textures, observed masses, family names, and likelihood preferences have no primitive authority. If such data decide which circuits are live, the decision is imported.
8. H_8 : *no imported evaluator*. Any additional residue gate must be standing-fixed by X_{gen} . If not, it is imported auxiliary data and fails same-scope admissibility by the auxiliary-datum dichotomy.

These eight clauses instantiate the fixed-domain comparison class locally. A proposed additional primitive orientation residue has only two same-scope forms. If it changes one of the fixed clauses, it

is a target update, extension, burden, or failed primitive. If it changes none of them while claiming primitive force, it is a standing-relevant discriminator/operator on X_{gen} . Fixed-domain exhaustion rules out a fifth governance-bearing locus, and same-scope operator closure classifies the resulting operator as bookkeeping-equivalent or inadmissible. $\square$

Corollary 8.2 (No untyped import into generation holonomy). *A proposed extra orientation-residue authorizer is same-scope only if it is well typed in $\mathcal{G}_{\text{hol}}$. Otherwise it is target update, extension, burden, or failed primitive.*

9 Raw, live, and primitive residue ranks

Definition 9.1 (Raw residue rank). *The raw transition residue module is*

$$\Omega_{\text{raw}}(\Gamma) = C^1(\Gamma)/dC^0(\Gamma), \quad (17)$$

where C^1 is the edge-coordinate group and dC^0 is the subgroup generated by vertex-local rephasings. For a connected graph, its rank is $\beta_1(\Gamma)$.

Definition 9.2 (Live residue rank). *The live residue module $\Omega_{\text{live}}(\Gamma) \subseteq \Omega_{\text{raw}}(\Gamma)$ consists of raw quotient residues that remain standing-relevant after admissible quotienting and no-selector normalization in $\mathcal{G}_{\text{hol}}$.*

Definition 9.3 (Primitive target rank). $\Omega_{\text{SM-gen}}^{\text{prim}}$ is the primitive same-scope generation target's CP/holonomy orientation-residue register. Its rank is the number of independent primitive closed transition-circuit residues the fixed target requires and licenses.

Theorem 9.4 (Role-incidence liveness theorem). *If $\text{ClosedGenCert}(n)$ holds, then every independent circuit of Γ_n is live for the generation-role certificate. Equivalently,*

$$\Omega_{\text{live}}(\Gamma_n) = \Omega_{\text{raw}}(\Gamma_n). \quad (18)$$

Proof. Let $\gamma \in \Omega_{\text{raw}}(\Gamma_n)$ be an independent raw circuit class. A closed generation-role certificate does not first count isolated vertices and then optionally decorate them with phase data. A generation role is a vertex as participating in the full closed transition-incidence structure. Likewise, an edge is not a fitted nonzero transition amplitude; it is a certified co-comparability channel of the common matter/mass/weak/flavor-transition target. Hence an independent circuit in Γ_n is not optional coordinate slack. It is an incidence relation among the same live role occupants whose vertices are being counted.

A registered phase coordinate may vary on an already-live circuit. Such a coordinate can describe, calibrate, or parameterize a live incidence relation after the circuit has standing. It cannot decide whether an independent circuit of the role-incidence graph is live in the first place. A rule that selects which independent circuits count toward role closure is a standing-relevant selector over the incidence structure, not a downstream coordinate value on an admitted circuit.

Suppose, toward contradiction, that γ is raw but not live. There are four exhaustive possibilities. First, γ collapses under the admissible quotient. Then it was not an independent element of $\Omega_{\text{raw}}(\Gamma_n)$. Second, γ is suppressed by a texture, zero setting, empirical fit, family label, basis convention, or chosen matrix form. Then the liveness decision is made by a hidden selector, contrary to the no-selector field of $\text{ClosedGenCert}(n)$. Third, γ is called a registered coordinate while its incidence is still used to retain the same generation-role occupants. Then the certificate counts the circuit's incidence for role closure while denying that incidence any standing authority. This is not ordinary

registered coordinate status; it is a hidden standing partition over the transition-incidence structure. Fixed-domain exhaustion sends such a partition to quotient collapse, boundary/role-typing change, standing-partition change, or continuation change. None is a same-scope closed certificate. Fourth, γ is removed by altering the flavor-transition target, the primitive residue gate, or the admissible continuation structure. Then the claim is an extension, target update, or burden rather than a closure of D_{gen} .

These alternatives exhaust the local loci supplied by fixed-domain exhaustion: quotient, standing partition, boundary/role typing, and admissible continuation. Since $\text{ClosedGenCert}(n)$ excludes quotient collapse of independent rank, hidden selector, registered-only role closure, and target change, γ must be live. Therefore every raw independent circuit is live and $\Omega_{\text{live}}(\Gamma_n) = \Omega_{\text{raw}}(\Gamma_n)$. $\square$

Corollary 9.5 (Live rank of a closed candidate). *If $\text{ClosedGenCert}(n)$, then*

$$\text{rank } \Omega_{\text{live}}(\Gamma_n) = \text{rank } \Omega_{\text{raw}}(\Gamma_n) = \beta_1(K_n). \quad (19)$$

10 Rank-one primitive target theorem

10.1 Lower bound: nonredundancy of the CP/holonomy carrier

A primitive carrier cannot be a merely decorative capacity. A carrier may license many registered capacities, possible coordinates, or downstream presentations; those capacities are not themselves primitive target roles. A primitive entry in C^{prim} , by contrast, must make a nonzero contribution to the fixed target's admissible consequence structure. If it has no occupied consequence, no forced target role, and no quotient-level standing effect, fixed-domain role exhaustiveness classifies it as skin, registered capacity, or bookkeeping rather than primitive standing.

Theorem 10.1 (Primitive carrier nonvacuity). *A lifted primitive carrier in the open-generation target must carry at least one admissible consequence or forced target role in D_{gen} . If it carries none, it is not primitive in D_{gen} ; it is skin, registered coordinate capacity, or bookkeeping.*

Proof. Fixed-domain role exhaustiveness classifies every standing-bearing component as anchor, tensor, or skin. A primitive carrier in a fixed target cannot be skin, because skin changes no admissible consequence and carries no standing authority. A merely empty capacity likewise contributes neither a target role nor a quotient-level consequence. Treating such a capacity as primitive would introduce a fourth status between skin and tensor, which role exhaustiveness disallows.

The lifted CP/holonomy carrier does not move the generation-holonomy boundary; it is not an anchor. It must therefore either contribute tensor-level orientation-residue standing or lose primitive status. Since it is received into D_{gen} as a primitive carrier-role requirement by Theorem 3.3, it must contribute at least one admissible consequence or forced target role. Nonvacuity follows. $\square$

Lemma 10.2 (CP/holonomy nonredundancy). *The primitive target rank is nonzero:*

$$\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} \geq 1. \quad (20)$$

Proof. The received SM flavor package has two primitive entries: flavor-transition standing and CP/holonomy-residue standing. Raw matrix entries, fitted angles, phase coordinates, basis choices, rephasings, and PMNS coordinates are registered statuses. Suppose $\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 0$. Then the CP/holonomy-residue carrier supplies no live primitive consequence distinct from flavor-transition standing or registered phase-coordinate status. By Theorem 3.3, the CP/holonomy carrier is

present in D_{gen} as a carrier-role requirement, not as a fixed-cardinality import. By Theorem 10.1, a primitive carrier in D_{gen} must carry at least one admissible consequence or forced target role. Fixed-domain role minimality classifies every standing-bearing component as anchor, tensor, or skin. The CP/holonomy carrier is not anchor, because the same flavor target remains defined without boundary motion. Under the rank-zero assumption it is not tensor, because it has no quotient-level primitive consequence. Therefore it is skin, registered coordinate capacity, or bookkeeping. That contradicts its lifted primitive-carrier role. Thus the rank is at least one. $\square$

10.2 Residue-to-operator bridge

The upper bound requires a local bridge: an independent primitive residue is not merely another value. It is a second standing-relevant classification of closed transition circuits.

Lemma 10.3 (Residue-to-operator lemma). *Let ω_1 be an admitted primitive orientation residue. If ω_2 is independent of ω_1 and is asserted as a primitive same-scope residue, then ω_2 induces a second same-scope standing-relevant operator/discriminator on X_{gen} .*

Proof. A primitive residue is not a numerical label. It affects the primitive standing partition for closed transition-circuit classes. Hence it defines a consequence functional

$$O_{\omega_2} : \Omega_{\text{raw}}(\Gamma_n) \longrightarrow \{\text{primitive, nonprimitive}\} \quad (21)$$

inside $\mathcal{G}_{\text{hol}}$. If O_{ω_2} factors through O_{ω_1} , then ω_2 is not independent primitive rank; it is bookkeeping-equivalent, quotient-duplicate, or registered coordinate status. If it does not factor through O_{ω_1} , it changes the primitive standing partition or adds a second primitive consequence slot. By Theorem 8.1, that is exactly a same-scope standing-relevant operator/discriminator. Therefore an independent primitive residue induces a second operator. $\square$

Lemma 10.4 (Auxiliary datum dichotomy for residue operators). *Every candidate second residue operator is either bookkeeping-equivalent to the existing primitive residue gate or inadmissible as a same-scope primitive operator.*

Proof. By the same-scope operator-closure source, a same-scope operator on a fixed standing-bearing carrier is governed by the auxiliary-datum dichotomy. If the datum needed to certify the second residue is already standing-fixed, the operator is bookkeeping-equivalent and adds no independent primitive rank. If the datum is not standing-fixed, it imports a selector, evaluator, witness, completion, comparison rule, or authority. Such import is inadmissible inside the fixed same-scope carrier. $\square$

Lemma 10.5 (No totalization or repair of residue standing). *A partial or absent residue certificate cannot be made primitive by adding a same-scope evaluator, selector, completion rule, limiting rule, canonical representative, or repair operator.*

Proof. Closure by exhaustion for same-scope operators rules out totalizing a partial operator on the standing-bearing carrier. Every totalization attempt is domain enlargement, carrier collapse, evaluative completion, conservative definitional extension, or non-conservative family change. In the generation-holonomy arena these are, respectively, newly defining old circuit certificates, changing the carrier, importing an evaluator or selector, bookkeeping, or target update. None yields a second same-scope primitive register. $\square$

Lemma 10.6 (No emergent second register). *No second primitive orientation-residue register emerges by composition, iteration, diagonalization, fixed-point detour, representation change, foundation swap, transfer, or interaction among registered flavor statuses.*

Proof. The same-scope operator closure theorem proves that no emergent operator family appears by assembly from closed families. Applying this locally, a putative second residue arising from combined CKM coordinates, PMNS boundary statuses, phase conventions, rephasings, endpoint phases, extension surfaces, or reconstruction routes is either a lawful redescription of existing standing or an inadmissible imported kernel. $\square$

10.3 Upper bound: no second primitive register

Proposition 10.7 (Second-residue branch exhaustion). *Let ω_2 be independent of an admitted primitive residue ω_1 . In the fixed same-scope generation target, ω_2 has exactly one of the following dispositions: quotient/bookkeeping duplicate; registered coordinate or boundary status; hidden selector or evaluator import; anchor motion or target update; tensor-load extension; burden, failed primitive, or hostile branch. It is not a second closed same-scope primitive register.*

Proof. If ω_2 is fixed by the existing standing quotient, it is bookkeeping by Lemma 10.4. If it changes no admissible consequence, role exhaustiveness classifies it as skin or registered status. If it requires a selector, evaluator, completion, canonical representative, or repair, Lemma 10.5 and Lemma 10.4 classify the move as imported and inadmissible. If it changes boundary typing or evaluated-fragment membership, it is anchor motion and therefore target update. If it changes quotient-level consequence by adding a new primitive slot, it is tensor-load extension or a new target component. If it is asserted without closing those obligations, it is burden or failed primitive. If all same-scope hostile obligations close, the theorem is falsified rather than protected. The fixed-domain exhaustion theorem states that no fifth governance-bearing locus remains, so the list is complete. $\square$

Lemma 10.8 (No surplus primitive register). *The same-scope Standard Model generation target licenses at most one independent primitive CP/holonomy residue register:*

$$\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} \leq 1. \quad (22)$$

Proof. Assume two independent primitive registers. By Lemma 10.3, the second induces a same-scope standing-relevant operator on X_{gen} . By Proposition 10.7, that operator is bookkeeping-equivalent, registered, inadmissible, target-updating, extension, burden, failed primitive, or hostile. None is a second closed primitive register inside the same target. Contradiction. $\square$

Theorem 10.9 (Rank-one primitive target theorem). *In the open-generation Standard Model carrier target,*

$$\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1. \quad (23)$$

Proof. The lower bound is Lemma 10.2. The upper bound is Lemma 10.8. Hence the rank is exactly one. The theorem does not say that all downstream phase coordinates vanish or that the carrier type cannot license registered statuses. It says that exactly one independent primitive orientation-residue register is available for same-scope generation-role closure. $\square$

Theorem 10.10 (Rank-matching condition). *If ClosedGenCert(n) holds, then*

$$\text{rank } \Omega_{\text{live}}(\Gamma_n) = \text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1. \quad (24)$$

Consequently,

$$\beta_1(K_n) = 1.$$

Proof. A closed generation certificate must realize the primitive target rank without underfilling or overfilling it. By Lemma 9.4, live rank equals raw rank in any closed certificate. By Corollary 7.5, raw rank equals $\beta_1(K_n)$. By Theorem 10.9, the primitive target rank is one. Hence (24) and $\beta_1(K_n) = 1$. $\square$

11 Neutrino boundary and PMNS phase boundary

11.1 Why this boundary is needed

The rank proof concerns primitive generation-transition residue. Neutrino boundary reconstruction, PMNS coordinates, oscillation baselines, and possible boundary phase statuses are not allowed to act as generation selectors. They also must not accidentally add primitive residue registers.

Definition 11.1 (Boundary phase status). *A neutrino-side phase or PMNS-like entry has boundary phase status when it is tied to source, propagation, detection, basis convention, mass/flavor eigenstate interface, matter route, or reconstruction kernel rather than to a same-scope primitive closed generation-transition circuit.*

Lemma 11.2 (PMNS/boundary phase nonpromotion). *PMNS coordinates, oscillation baselines, weak-flavor boundary labels, Majorana-like boundary phases, and neutrino mass-ordering surfaces do not increase $\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}}$ unless they close an independent same-scope primitive CP/holonomy-residue certificate.*

Proof. The neutrino reconstruction sources treat flavor under noncommuting propagation as boundary reconstruction, not as globally transported primitive flavor identity. NNR6 supplies PMNS-like interface discipline and hidden flavor-transport exclusion. Therefore neutrino-side coordinates and phases are registered boundary, route, mixing, projection, or burden statuses unless a primitive holonomy certificate closes. If such a certificate does close as an additional independent primitive register, it triggers the surplus-register branch of Lemma 10.8; it is not silently absorbed as an empirical selector. $\square$

Corollary 11.3 (No neutrino selector leak). *Neutrino boundary reconstruction supports $C_{\text{boundary}}(n)$ but does not supply an independent proof of $n = 3$ from observed flavor count, PMNS dimension, oscillation baselines, or boundary phase coordinate count.*

12 Lower-cardinality eliminators

Theorem 12.1 (No zero-generation closure).

$$\neg \text{ClosedGenCert}(0). \quad (25)$$

Proof. For $n = 0$, there is no candidate generation-role occupant, no live generation residue, and no transition graph. Hence no role-covering assignment exists. In particular, the representation and flavor-transition components fail. Therefore $\text{ClosedGenCert}(0)$ does not close. $\square$

Theorem 12.2 (No one-generation closure).

$$\neg \text{ClosedGenCert}(1). \quad (26)$$

Proof. A one-role presentation may support local matter-response standing, charge compatibility, weak response, and mass-response-slot status. It does not support nontrivial inter-role flavor-transition standing. Equivalently, K_1 has no edge and no circuit. Thus the flavor-transition and holonomy-residue certificate fields fail. $\square$

Theorem 12.3 (No two-generation closure).

$$\neg \text{ClosedGenCert}(2). \quad (27)$$

Proof. Assume $\text{ClosedGenCert}(2)$. Then $\Gamma_2 \simeq K_2$, and $\beta_1(K_2) = 0$. By Theorem 10.10, any closed certificate must have $\beta_1(K_n) = 1$. Contradiction. Thus two-role systems can have a comparison edge, but after rephasing quotient they have no closed transition-circuit residue matching the primitive SM generation target. $\square$

13 Positive closure at $n = 3$

13.1 Target role set of the minimal closed circuit

The positive direction must not introduce empirical family labels. The three roles are structural positions in the minimal closed transition circuit.

Definition 13.1 (Minimal circuit incidence positions). *Let*

$$\mathcal{I}_3 := V(K_3)/\text{Aut}(K_3) \cong V(K_3)/S_3$$

denote the unordered incidence-position set of the minimal closed generation-transition circuit. We may write three representatives as

$$R_A, \quad R_T, \quad R_H,$$

when a displayed assignment requires names. These symbols are expository gauges for incidence positions, not qualitative family roles and not empirical labels. The official target-role object is $V(K_3)/S_3$.

Lemma 13.2 (Rank-one circuit incidence). *A nonzero primitive closed transition-circuit residue with rank one forces exactly three incidence positions in the minimal same-scope generation target.*

Proof. A simple closed circuit requires at least three vertices. One or two vertices have cycle rank zero and cannot host a primitive closed transition-circuit residue. A triangle has exactly one independent circuit and supplies exactly the three incidence positions needed by a rank-one primitive circuit. Any additional live incidence position in the complete co-comparability normal form raises the raw and live cycle rank above one by Lemma 9.4; by Theorem 10.9 that is surplus primitive rank unless the candidate quotient-collapses or exits the same-scope target. Therefore the minimal and saturating incidence support of the rank-one primitive target is precisely $\mathcal{I}_3$. $\square$

Theorem 13.3 (Target role set forcing at rank one). *If the open-generation target has primitive holonomy rank one, then*

$$\text{RT}(D_{\text{gen}}, \text{GenerationRole}) = \mathcal{I}_3 = V(K_3)/S_3. \quad (28)$$

The symbols R_A, R_T, R_H are only named representatives of these three incidence positions.

Proof. By Theorem 10.9, the open-generation target has exactly one primitive closed transition-circuit register. By Lemma 13.2, that register forces exactly three incidence positions in the minimal same-scope target. These positions are roles only because a closed certificate must occupy the incidence positions by eligible residues; they are not empirical labels and not three distinct qualitative species. The presentation quotient deletes family names, field copies, basis choices, rephasings, matrix coordinates, and mass-value displays. Fewer incidence positions under-occupy the primitive circuit. More incidence positions force surplus live circuit rank by complete co-comparability and Lemma 9.4. Thus the forced target role set is the unordered three-position incidence set $\mathcal{I}_3 = V(K_3)/S_3$. Any use of R_A, R_T, R_H below is a notational choice of representatives, not an added qualitative role taxonomy. $\square$

Lemma 13.4 (Eligibility of the three abstract residues). *There exist three presentation-quotiented live residues $a, t, h \in \overline{\text{Residue}}_3^{\text{live}}$ such that*

$$\text{EC}(a, R_A), \quad \text{EC}(t, R_T), \quad \text{EC}(h, R_H).$$

Proof. The open-generation arena preserves the matter carrier, charge/weak response, mass-response slot, flavor-transition carrier, boundary-reconstruction interface, and rank-one CP/holonomy target. By Theorem 13.3, the target roles are exactly the three incidence positions of $\mathcal{I}_3$. A three-residue presentation supplies exactly three live vertices for the minimal closed transition circuit. After quotienting presentation labels, each live residue is eligible for one incidence position; no residue is selected by a family name, matrix row, or mass value. $\square$

Lemma 13.5 (Joint compatibility of the triangle). *The assignment*

$$\sigma_3(R_A) = a, \quad \sigma_3(R_T) = t, \quad \sigma_3(R_H) = h$$

has $\text{JC}(\sigma_3) = 1$.

Proof. The graph on a, t, h is K_3 , so it has exactly one closed transition circuit. Representation, charge, weak, mass-response, flavor-transition, and boundary compatibility are inherited from D_{gen} . Vertex-local phase skins are removed by rephasing quotient. The remaining circuit supplies exactly one live CP/holonomy residue, matching rank $\Omega_{\text{SM-gen}}^{\text{prim}} = 1$. There is no surplus residue, no hidden selector, and no hidden load. Therefore the assignment is jointly compatible. $\square$

Lemma 13.6 (Uniqueness up to role-incidence automorphism). *Any closed three-role assignment is equivalent to σ_3 up to the automorphism group of the triangular role-incidence graph, i.e. up to S_3 permutation and presentation quotient.*

Proof. Any closed three-role assignment has transition graph K_3 and one independent circuit. Each vertex is necessary for the circuit; deleting any one vertex gives cycle rank zero. Since empirical labels, basis choices, rephasings, field-copy surfaces, matrix coordinates, and mass-value displays are quotiented, the remaining freedom is exactly permutation of the three incidence positions. Hence all closed assignments lie in the same role-occupancy orbit. $\square$

Theorem 13.7 (Three-role positive certificate).

$$\text{ClosedGenCert}(3). \tag{29}$$

Proof. The forced target role set is $\mathcal{I}_3 = V(K_3)/S_3$ by Theorem 13.3. Pointwise eligibility follows from Lemma 13.4. Joint compatibility follows from Lemma 13.5. Uniqueness follows from Lemma 13.6. No empirical label, matrix dimension, mass table, or field-copy selector is used. Therefore $\text{ClosedGenCert}(3)$ holds. $\square$

14 Higher-cardinality eliminators

14.1 Surplus circuit pressure

For $n > 3$, the closed co-comparability normal form K_n has

$$\beta_1(K_n) = \frac{(n-1)(n-2)}{2} > 1.$$

Thus a closed higher-cardinality candidate would carry more independent non-removable transition-circuit residues than the same-scope target licenses.

Definition 14.1 (Surplus residue). *A candidate n -presentation has surplus transition-holonomy residue when $\text{rank } \Omega_{\text{live}}(\Gamma_n) > \text{rank } \Omega_{\text{SM-gen}}^{\text{prim}}$ after presentation quotient and rephasing quotient.*

Lemma 14.2 (Surplus-residue alternatives). *For $n > 3$, a candidate can avoid surplus primitive residue only by one of the following dispositions:*

1. *quotient collapse to a lower live role count;*
2. *hidden selector or texture suppressing extra circuits;*
3. *hidden load adding new primitive constraints, source structure, or registers;*
4. *extension disposition;*
5. *target update;*
6. *incompatible occupancy;*
7. *unresolved hostile role-occupancy burden.*

Proof. Because $\beta_1(K_n) > 1$, the candidate has extra independent cycle-space generators. If those generators are not live after quotient, the live role count collapses. If they are declared zero, tied, or inert by a rule not forced by the role certificate, a hidden selector or texture has been imported. If a new mechanism controls them, the claim has added source, flavor, Higgs, representation, boundary, or primitive residue load. Such addition is an extension or target update unless it closes a same-scope hostile certificate. If the assignment cannot make the extra generators compatible with the single target register, joint compatibility fails. If the claim asserts closure without resolving these alternatives, it remains a burden. $\square$

Theorem 14.3 (No higher-generation closure). *For all $n > 3$,*

$$\neg \text{ClosedGenCert}(n). \tag{30}$$

Proof. Assume $n > 3$ and $\text{ClosedGenCert}(n)$. By Theorem 10.10, $\beta_1(K_n) = 1$. But for $n > 3$, $\beta_1(K_n) > 1$. Contradiction. Equivalently, the extra circuit rank must be disposed of by one of Lemma 14.2; each disposition blocks closed same-scope $n > 3$ generation occupancy. Therefore no higher-cardinality candidate closes. $\square$

Remark 14.4. The theorem grants that a fourth field copy, anomaly-neutral family repetition, or 4×4 mixing matrix can be written. Syntax is not role occupancy. A same-scope higher-generation claim must close the no-surplus rank match; under D_{gen} , it does not close as a same-scope generation-role certificate.

15 Main theorem

Theorem 15.1 (Generation cardinality as forced role-occupancy endpoint). *For every $n \in \mathbb{N}$,*

$$\text{ClosedGenCert}(n) \iff n = 3. \quad (31)$$

Proof. If $n = 3$, Theorem 13.7 gives $\text{ClosedGenCert}(3)$. If $n = 0$, Theorem 12.1 blocks closure; if $n = 1$, Theorem 12.2 blocks closure; if $n = 2$, Theorem 12.3 blocks closure; and if $n > 3$, Theorem 14.3 blocks closure. These cases exhaust $\mathbb{N}$. Hence $\text{ClosedGenCert}(n)$ holds exactly when $n = 3$. $\square$

Corollary 15.2 (Target role-cardinality).

$$|\text{RT}(D_{\text{gen}}, \text{GenerationRole})| = 3. \quad (32)$$

Corollary 15.3 (Quotient residue cardinality). *For a closed same-scope generation certificate,*

$$|\overline{\text{Residue}}_{\text{gen}}^{\text{live}}| = 3. \quad (33)$$

Remark 15.4. These cardinality equations are corollaries. They are not obtained from observed family labels, matrix dimension, or field-copy syntax.

16 Neutrino, PMNS, and endpoint-sensitive phase audit

16.1 Boundary reconstruction is not hidden flavor transport

Neutrino flavor labels remain boundary/projector witnesses unless propagation preserves the flavor projector algebra through the interior. The neutrino reconstruction source and NNR6 block hidden global flavor transport. PMNS-like standing attaches to a quotient-stable boundary transition carrier, not raw PMNS matrix presentation.

Theorem 16.1 (PMNS boundary-status separation). *PMNS entries, weak flavor labels, oscillation baselines, mass-splitting routes, endpoint-sensitive phases, and Majorana-like phase surfaces do not add primitive generation holonomy-rank unless they close the same CP/holonomy-residue certificate.*

Proof. Neutrino reconstruction classifies PMNS-like material as boundary reconstruction, transition-kernel, route, projection, or witness status. NNR6 states that the boundary carrier may be licensed while raw PMNS import, hidden flavor transport, mixing-angle value extraction, and CP-phase value extraction remain blocked. Therefore neutrino phase surfaces are either registered boundary statuses, endpoint-sensitive registered phases, burdens, extensions, target updates, or hostile certificate attempts. They do not silently supply a second primitive $\Omega_{\text{SM-gen}}^{\text{prim}}$ register. $\square$

17 Fourth-generation hostile reconstruction

Protocol 17.1 (Fourth-generation audit). *A fourth-generation claim f must pass the following sequence.*

1. *Same-scope test: does f preserve D_{gen} and the primitive carrier target?*
2. *Presentation test: does f supply only field duplication, family label, mass values, anomaly repetition, or matrix enlargement?*

3. Rank test: does f avoid increasing $\text{rank } \Omega_{\text{live}}(K_n)$ above one?
4. Selector test: does f avoid texture, zero-setting, fit, basis, or empirical suppression as hidden selector?
5. Load test: does f avoid new Higgs, fermion, representation, boundary, or flavor-transition load?
6. Occupancy test: does f close EC, JC, and unique compatible role occupancy?

Theorem 17.2 (Fourth-generation branch). *A fourth-generation claim is classified as quotient collapse, failed primitive duplication, role-cardinality burden, registered extension, target update, hidden-load violation, hidden-selector violation, or hostile role-occupancy branch. It is not a closed same-scope generation certificate.*

Proof. A fourth candidate has $n = 4$, hence $\beta_1(K_4) = 3$. Since $\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1$, rank matching fails unless the extra residue is removed. All lawful removal routes are quotient collapse, registered coordinate status, burden, extension, target update, hidden selector/load failure, or hostile certificate. None yields $\text{ClosedGenCert}(4)$. $\square$

Remark 17.3. The proof does not deny that a fourth-generation field-copy model can be written. It denies that field-copy syntax plus enlarged mixing matrices closes the same-scope open-generation role-occupancy certificate.

18 Conditional reduction under withheld rank-one target

Theorem 18.1 (Conditional reduction theorem). *Even if Theorem 10.9 is withheld as a received/closed premise, the following conditional theorem is closed:*

$$\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1 \quad \Rightarrow \quad (\text{ClosedGenCert}(n) \iff n = 3). \quad (34)$$

Proof. Assume $\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1$. The complete graph normal form, rephasing-cycle quotient, and rank-matching theorem do not depend on the proof of the rank-one target theorem; they only use its conclusion. Therefore $\text{ClosedGenCert}(n) \Rightarrow \beta_1(K_n) = 1 \Rightarrow n = 3$, and Theorem 13.7 supplies the reverse implication. $\square$

Audit note 18.2. This conditional corollary records the exact residual dependency. A reader who withholds the rank-one primitive target theorem can still evaluate the independent reduction from rank one to three-generation closure.

19 Referee objection lock

Table 4: Hostile-referee objections and responses.

Objection	Answer
This assumes three generations.	The proof begins in D_{gen} , not $D_{\text{flavor}}^{N_g}$. Cardinality follows from rank matching.
One carrier type can host many residues.	Agreed. The proof separately proves $\text{rank } \Omega_{\text{SM-gen}}^{\text{prim}} = 1$ as a primitive target-rank theorem.

Objection	Answer
The graph argument is metaphorical.	The proof identifies edge phases modulo vertex rephasings with the cycle-space quotient $C^1/\text{im } d$.
Why complete graph rather than connected graph?	Γ_n is the graph of direct co-comparability slots, not a route graph. Closed generation-role standing requires pairwise co-comparability. A missing pairwise co-comparability slot is a certificate failure, not a zero transition amplitude.
A fourth-generation model can be written.	Field-copy syntax and 4×4 matrices do not close role occupancy. K_4 has rank three and violates rank matching unless it collapses or changes status.
CP violation requiring three generations is empirical.	The proof does not use observed CP violation. It uses structural CP/holonomy-residue rank after rephasing quotient.
PMNS or Majorana phases add extra phases.	They are boundary, endpoint-sensitive, registered, burden, or target-update statuses unless they close the same primitive holonomy certificate.
Anomaly cancellation does not force three.	Correct. Anomaly discipline is a guard, not the proof engine.

20 Future formalization targets

This section records future Lean-facing targets. It is not used as formal support for the present manuscript; the proofs above are the manuscript's load-bearing content. These declarations are a dependency map, not an independent formal verification. A later audit should replace the displayed axioms by proved graph-theoretic and role-occupancy lemmas.

```

constant ClosedGenCert : Nat -> Prop
constant TransitionGraph : Nat -> Type
constant CompleteGraph : Nat -> Type
constant OmegaRawRank : Nat -> Nat
constant OmegaLiveRank : Nat -> Nat
constant OmegaPrimRank : Nat

axiom complete_graph_normal_form :
  ClosedGenCert n -> TransitionGraph n = CompleteGraph n

axiom rephasing_cycle_rank :
  OmegaRawRank n = ((n - 1) * (n - 2)) / 2

theorem rank_one_target_by_fixed_domain_exhaustion :
  OmegaPrimRank = 1

theorem rank_match :
  ClosedGenCert n -> OmegaLiveRank n = OmegaPrimRank

```



```
theorem closed_gen_cert_implies_three :
  ClosedGenCert n -> n = 3
```

A fuller implementation should replace the displayed axioms by graph-theory proofs for $\beta_1(K_n)$, a formal definition of live residue rank, a mechanized role-occupancy certificate, and uniqueness modulo the role-incidence automorphism group S_3 .

21 Conclusion

The proof has a clean core. The cumulative Standard Model carrier constraints, together with fixed-domain exhaustion and same-scope operator closure, force a rank target: one primitive CP/holonomy generation-transition register. A closed generation certificate forces a complete co-comparability graph. Rephasing quotienting identifies independent phase/holonomy residue slots with the cycle rank of that graph. The complete graph rank is

$$\beta_1(K_n) = \frac{(n-1)(n-2)}{2}.$$

Rank matching with the single primitive target gives $n = 3$, and the unordered triangle incidence assignment closes the constructive direction. Thus generation cardinality is not read from empirical families or matrix size. It is the unique same-scope role-occupancy endpoint of the Standard Model carrier architecture. Target-updating or extension theories may contain additional fermion-family structure, but such additions do not close the same-scope Standard Model generation-role certificate. A valid same-scope hostile certificate with different primitive residue rank would falsify the theorem rather than be reclassified away.

A Revision integration ledger

Item	Integrated requirement	Manuscript location
1	One carrier type versus one primitive register	Section 10
2	Rephasing quotient as cycle residue	Section 7
3	Complete graph normal form	Section 6
4	Construct $n = 3$ assignment	Section 13
5	Strong upper-bound theorem	Section 14
6	Neutrino/PMNS phase status	Section 16
7	Theorem-level traceability	Table 1
8	Open-arena CP/holonomy lifting without fixed N_g import	Theorem 3.3
9	Local fixed-domain instantiation checklist	Table 3
10	Role-incidence liveness theorem	Lemma 9.4

11	Final minor referee hardening: primitive-carrier nonvacuity wording, cross-reference labels, same-scope boundary preservation, and formalization-section modesty	Theorem 10.1 and section 20 and conclusion
12	Final technical correction: explicit cleveref environment names and complete-graph wording hardening	preamble, Table 4

B Cardinality audit table

n	$\beta_1(K_n)$	Certificate status
0	none/0	no generation-role occupancy
1	0	no inter-role transition
2	0	transition edge but no closed holonomy circuit
3	1	exact rank match; closed
4	3	surplus residue rank
> 4	> 3	surplus residue rank